

MAT-124 Advanced Mathematics I

Control 2 — Full Solutions & Grading Rubric

May 22, 2026 • 70 minutes • 60 points

Notes for graders.

- Each question includes the full worked solution, a graphical representation of the solution set, a point-by-point rubric, and coherence rules for error propagation.
- **Coherence principle:** a mistake is penalized *once*. If the student reasons consistently after an error, subsequent points are preserved.
- Alternative valid methods are accepted if mathematically sound.
- Maximum-score criteria are stated explicitly for each question.
- **Key pedagogical distinction:** Questions 1 and 2 produce sets with no minimum and no maximum (open intervals). Question 3 produces a set with a minimum but no maximum (half-open interval $[40, 50)$). Students who fail to distinguish these lose significant points in part (c) of Q3.

Question	Topic	Solution set	Max pts
1	$ x^2 - 4 < 3(x + 2)$	$A = (-1, 5)$ — no min, no max	30
2	$R(x) > C(x)$ mining	$A = (10, 40)$ — no min, no max	30
3	Supply & demand	$N = [40, 50)$ — min exists, no max	30
Total			60

Question 1

(30 points)

Statement. Consider the set $A = \{x \in \mathbb{R} : |x^2 - 4| < 3(x + 2)\}$.

Part a

(18 pts)

Determine A in interval notation.

Step 1 — Necessary condition.

The right-hand side must be strictly positive for the inequality to hold (since $|x^2 - 4| \geq 0$). Therefore:

$$3(x + 2) > 0 \implies x > -2$$

Any solution must satisfy $x > -2$. We restrict our analysis to $x > -2$.

Step 2 — Identify critical points and cases.

Factor: $x^2 - 4 = (x - 2)(x + 2)$. The expression changes sign at $x = -2$ and $x = 2$.

Combined with the domain $x > -2$, the two cases are:

- **Case 1:** $-2 < x \leq 2 \implies |x^2 - 4| = 4 - x^2$
- **Case 2:** $x > 2 \implies |x^2 - 4| = x^2 - 4$

Case 1: $-2 < x \leq 2$. The inequality becomes $4 - x^2 < 3(x + 2)$.

$$4 - x^2 < 3x + 6 \implies 0 < x^2 + 3x + 2 \implies 0 < (x + 1)(x + 2)$$

The critical points of $(x + 1)(x + 2)$ are $x = -2$ and $x = -1$. Sign table on $(-2, 2]$:

x	$(-2, -1)$	-1	$(-1, 2]$
$(x + 1)$	$-$	0	$+$
$(x + 2)$	$+$	$+$	$+$
$(x + 1)(x + 2)$	$-$	0	$+$
$0 < (x + 1)(x + 2)?$	No	No	Yes

Solution from Case 1: $x \in (-1, 2]$

Case 2: $x > 2$. The inequality becomes $x^2 - 4 < 3(x + 2)$.

$$x^2 - 4 < 3x + 6 \implies x^2 - 3x - 10 < 0 \implies (x - 5)(x + 2) < 0$$

The critical points of $(x - 5)(x + 2)$ are $x = -2$ and $x = 5$. Sign table on $(2, +\infty)$:

x	$(2, 5)$	5	$(5, +\infty)$
$(x - 5)$	$-$	0	$+$
$(x + 2)$	$+$	$+$	$+$
$(x - 5)(x + 2)$	$-$	0	$+$
$(x - 5)(x + 2) < 0?$	Yes	No	No

Solution from Case 2: $x \in (2, 5)$

Step 3 — Global sign table and combination.

Putting both cases together over the full domain $x > -2$:

x	$(-2, -1)$	-1	$(-1, 2]$	$(2, 5)$	5	$(5, +\infty)$
Case	1	1	1	2	2	2
Inequality satisfied?	No	No	Yes	Yes	No	No

$$A = (-1, 2] \cup (2, 5) = \boxed{(-1, 5)}$$

The union is a single open interval because the two pieces meet seamlessly at $x = 2$ (which belongs to Case 1 but not Case 2, and overall the transition is continuous).

Part b

(12 pts)

Determine $\inf A$, $\sup A$, $\min$, $\max$.

$$A = (-1, 5).$$

Set of lower bounds of A .

A number m is a lower bound of A if $m \leq x$ for all $x \in A$. Since every $x \in A$ satisfies $x > -1$, any $m \leq -1$ is a lower bound. No number $m > -1$ is a lower bound, since m itself (or any x with $-1 < x < m$) belongs to A and violates the condition.

$$\text{Set of lower bounds of } A = (-\infty, -1]$$

$$\inf A = -1.$$

The infimum is the greatest element of the set of lower bounds: $\inf A = \max(-\infty, -1] = -1$.

Does $\min A$ exist? The minimum must be the smallest element of A itself. Since $-1 \notin A$ (at

$x = -1$: $(x+1)(x+2) = 0$, the strict inequality fails), A has **no minimum**.

Set of upper bounds of A .

A number M is an upper bound of A if $M \geq x$ for all $x \in A$. Since every $x \in A$ satisfies $x < 5$, any $M \geq 5$ is an upper bound. No number $M < 5$ is an upper bound, since there exist elements of A arbitrarily close to 5.

$$\text{Set of upper bounds of } A = [5, +\infty)$$

$$\sup A = 5.$$

The supremum is the smallest element of the set of upper bounds: $\sup A = \min[5, +\infty) = 5$.

Does $\max A$ exist? Since $5 \notin A$ (at $x = 5$: $(x-5)(x+2) = 0$, the strict inequality fails), A has **no maximum**.

$$\boxed{\inf A = -1 \text{ (no min)} \quad \sup A = 5 \text{ (no max)}}$$

Rubric — Question 1

Criterion	Pts
Identifies necessary condition $x > -2$ (or equivalent domain argument)	3
Correctly handles Case 1 ($-2 < x \leq 2$): removes absolute value, factors, solves	4
Correctly handles Case 2 ($x > 2$): removes absolute value, factors, solves	4
Combines both cases correctly to obtain $A = (-1, 5)$	3
$\inf A = -1$ with formal justification (set of lower bounds)	3
$-1 \notin A$ argued correctly (min does not exist)	3
$\sup A = 5$ with formal justification (set of upper bounds)	3
$5 \notin A$ argued correctly (max does not exist)	3
Total	30

Full marks require: both cases with domain restriction; union producing $(-1, 5)$; all four claims in (b) formally justified with endpoint verification.

Coherence rules — Question 1

- **Omits $x > -2$:** no penalty if final answer is $(-1, 5)$. Deduct 2 pts only if the omission produces a wrong domain.
- **Error in one case:** deduct 4 pts for that case. Award 2 combination pts if the student unions correctly.
- **Writes $A = [-1, 5]$ or similar:** deduct 2 pts. Grade (b) relative to student's interval.
- **Does not verify endpoint non-membership in (b):** deduct 2 pts per missing verification.
- **Confuses $\inf$ with $\min$ or $\sup$ with $\max$:** deduct 2 pts total.

Question 2

(30 points)

Statement. $R(x) = 4.00x$, $C(x) = 0.05x^2 + 1.50x + 20$, both in millions of USD, $x \geq 0$.

Part a

(18 pts)

Find all $x \geq 0$ with $R(x) > C(x)$.

Step 1 — Set up the inequality.

$$4.00x > 0.05x^2 + 1.50x + 20$$

$$0 > 0.05x^2 - 2.50x + 20$$

Multiply both sides by -20 (reversing the inequality):

$$x^2 - 50x + 400 < 0$$

Step 2 — Factor.

We need two numbers that multiply to 400 and sum to 50: those are 10 and 40.

$$(x - 10)(x - 40) < 0$$

A product of two factors is negative when exactly one factor is negative, i.e., when $10 < x < 40$.

Step 3 — Apply domain $x \geq 0$.

$(10, 40) \subset [0, +\infty)$, so the domain adds no restriction.

$$x \in (10, 40)$$

Economic interpretation: the mine must produce more than 10 million pounds to overcome the fixed tax burden and break even. Beyond 40 million pounds, the exponentially growing fuel cost from ore grade depletion consumes all revenue. The viable window is exactly $(10, 40)$.

Part b

(12 pts)

Let $A = \{x \geq 0 : R(x) > C(x)\}$. Find $\inf A$, $\sup A$, $\min$, $\max$.

$$A = (10, 40).$$

Set of lower bounds of A .

Every $x \in A$ satisfies $x > 10$, so any $m \leq 10$ is a lower bound. No $m > 10$ is a lower bound since

elements of A exist between 10 and m .

Set of lower bounds of $A = (-\infty, 10]$

$\inf A = 10$.

$\inf A = \max(-\infty, 10] = 10$.

Does $\min A$ exist? $10 \notin A$: $R(10) - C(10) = 4(10) - (0.05(100) + 1.5(10) + 20) = 40 - 40 = 0$. At $x = 10$, profit is exactly zero — not strictly positive. **No minimum.**

Interpretation: the mine must produce strictly more than 10 M lbs to begin covering the transitory tax.

Set of upper bounds of A .

Every $x \in A$ satisfies $x < 40$, so any $M \geq 40$ is an upper bound. No $M < 40$ is an upper bound since elements of A exist between M and 40.

Set of upper bounds of $A = [40, +\infty)$

$\sup A = 40$.

$\sup A = \min[40, +\infty) = 40$.

Does $\max A$ exist? $40 \notin A$: $R(40) - C(40) = 4(40) - (0.05(1600) + 1.5(40) + 20) = 160 - 160 = 0$. Break-even again. **No maximum.**

Interpretation: ore depletion costs exactly equal revenue at 40 M lbs — the mine must stay strictly below this threshold.

$\inf A = 10$ (no min)	$\sup A = 40$ (no max)
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Rubric — Question 2

Criterion	Pts
Sets up inequality $R(x) > C(x)$ and rearranges correctly	3
Reduces to $(x - 10)(x - 40) < 0$ or equivalent	4
States $A = (10, 40)$ with domain $x \geq 0$ applied	3
Absent: no partial credit for setup only	—
$\inf A = 10$ with formal justification	3
$10 \notin A$ verified (min does not exist) + interpretation	3
$\sup A = 40$ with formal justification	3
$40 \notin A$ verified (max does not exist) + interpretation	3
Total	30

Coherence rules — Question 2

- **Arithmetic error in rearrangement:** deduct setup 3 pts. Award factorization and interval pts if consistent.
- **Uses quadratic formula instead of factoring:** full credit — any correct method accepted.
- **Writes $A = [10, 40]$:** deduct 3 pts (setup) + 2 pts per wrong endpoint claim in (b).
- **Verifies $R(10) = C(10)$ or $R(40) = C(40)$ but does not state the conclusion:** deduct 1 pt per missing conclusion.
- **Omits economic interpretations:** deduct 1 pt per missing sentence.
- **Does not apply domain $x \geq 0$:** no penalty if $(10, 40) \subset [0, +\infty)$ makes the restriction irrelevant (which it does here).

Question 3

(30 points)

Statement. $D(x) = 6143.6 - 99.9x$, $S(x) = 1148.8$, $x \geq 0$. Intervention when $D(x) - S(x) > M > 0$.

Part a

(6 pts)

Verify equilibrium. Find shortage and surplus zones.

Equilibrium verification.

$$D(50) = 6143.6 - 99.9(50) = 6143.6 - 4995 = 1148.6 \approx 1148.8 = S(50) \checkmark$$

(The small discrepancy arises from rounding the coefficient $99.9 \approx 4994.8/50$; the model is calibrated to the empirical elasticity $\varepsilon_d = -0.23$.)

Shortage zone ($D(x) > S(x)$).

$$6143.6 - 99.9x > 1148.8 \implies 4994.8 > 99.9x \implies x < \frac{4994.8}{99.9} \approx 50$$

Shortage zone: $[0, 50)$.

Surplus zone ($D(x) < S(x)$).

$$x > 50$$

Surplus zone: $(50, +\infty)$.

Economic reading: below 50 M liters/week, Santiago consumers would pay more than the current price — the market is undersupplied. Above 50 M liters/week, supply exceeds what consumers are willing to pay at current price levels.

Part b

(14 pts)

Find $I(M)$ and $N(M)$ in terms of M . When is $N(M)$ non-empty? Find M with $|N(M)| = 10$.

Intervention zone $I(M)$.

$$D(x) - S(x) > M \implies 4994.8 - 99.9x > M \implies x < \frac{4994.8 - M}{99.9}$$

$$I(M) = \left[0, \frac{4994.8 - M}{99.9} \right)$$

Non-intervention shortage zone $N(M)$.

The shortage zone is $[0, 50)$. Removing $I(M)$:

$$N(M) = \left[\frac{4994.8 - M}{99.9}, 50 \right)$$

When is $N(M)$ non-empty?

$N(M)$ is non-empty when its left endpoint is strictly less than 50:

$$\frac{4994.8 - M}{99.9} < 50 \implies 4994.8 - M < 4995 \implies M > -0.2$$

Since $M > 0$ by assumption, this is always satisfied. Also, $I(M)$ must be non-empty (otherwise intervention is never triggered), requiring $\frac{4994.8 - M}{99.9} > 0$, i.e., $M < 4994.8$.

$N(M)$ is **non-empty for all** $M \in (0, 4994.8)$.

Find M such that $|N(M)| = 10$.

$$|N(M)| = 50 - \frac{4994.8 - M}{99.9} = 10$$

$$\frac{4994.8 - M}{99.9} = 40 \implies 4994.8 - M = 3996 \implies \boxed{M = 998.8 \text{ CLP/liter}}$$

Note: $M = 998.8$ is approximately 87% of the pre-intervention price $P^* = 1148.8$ — the government only intervenes when the shortage magnitude exceeds the equivalent of nearly the full pre-crisis price per liter.

Part c

(10 pts)

With $M = 998.8$: find $\inf N$, $\sup N$, **min, **max**.**

With $M = 998.8$:

$$N(998.8) = \left[\frac{4994.8 - 998.8}{99.9}, 50 \right) = \left[\frac{3996}{99.9}, 50 \right) = [40, 50)$$

Set of lower bounds of N .

Every $x \in N$ satisfies $x \geq 40$, so any $m \leq 40$ is a lower bound. No $m > 40$ is a lower bound since $40 \in N$ and $40 < m$.

$$\text{Set of lower bounds of } N = (-\infty, 40]$$

$\inf N = 40$ and $\min N = 40$ exists.

$$\inf N = \max(-\infty, 40] = 40.$$

Does $\min N$ exist? We must check whether $40 \in N$. At $x = 40$:

$$D(40) - S(40) = 4994.8 - 99.9(40) = 4994.8 - 3996 = 998.8 = M$$

The condition for $I(M)$ requires $D(x) - S(x) > \mathbf{M}$ (strict). At $x = 40$ the shortage equals M but does not exceed it, so $40 \notin I(M)$. Also $D(40) = 2147.6 > 1148.8 = S(40)$, so there is a shortage. Therefore $40 \in N$.

Since $40 \in N$ and 40 is the infimum, $\min N = 40$ **exists**. (*This is the crucial difference from Q1 and Q2.*)

Interpretation: there is a smallest production level (40 M liters/week) at which the market has a shortage but the shortage exactly equals the intervention threshold — the government does not yet intervene.

Set of upper bounds of N .

Every $x \in N$ satisfies $x < 50$, so any $M \geq 50$ is an upper bound. No $M < 50$ is an upper bound since elements of N exist arbitrarily close to 50.

$$\text{Set of upper bounds of } N = [50, +\infty)$$

$\sup N = 50$ (**no maximum**).

$$\sup N = \min[50, +\infty) = 50.$$

Does $\max N$ exist? $50 \notin N$: at $x = 50$, $D(50) = S(50)$ — equilibrium, no shortage. **No maximum.**

Interpretation: the non-intervention shortage zone approaches but never reaches the equilibrium quantity — as consumption nears 50 M liters/week, the shortage dissolves and the market self-corrects.

$\min N = \inf N = 40 \qquad \sup N = 50 \text{ (no maximum)}$
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Rubric — Question 3

Criterion	Pts
Verifies $D(50) \approx S(50) = 1148.8$	3
Correctly identifies shortage zone $[0, 50)$ and surplus zone $(50, +\infty)$	6
Economic reading of shortage/surplus	5
Correctly derives $I(M)$ in terms of M	4
Correctly derives $N(M)$ in terms of M	4
Correctly determines $N(M)$ non-empty for $M \in (0, 4994.8)$	2
Finds $M = 998.8$ from length condition	4
$\inf N = 40$ with formal justification	3
$\min N = 40$ exists — formally verified $40 \in N$	2
Economic interpretation of $\inf N$	2
$\sup N = 50$ with formal justification	1
$\max N$ does not exist — formally verified $50 \notin N$	1
Economic interpretation of $\sup N$	1
Total	30

Full marks require: (i) verified equilibrium; (ii) both zones; (iii) $I(M)$ and $N(M)$ in terms of M ; (iv) non-emptiness condition; (v) $M = 998.8$; (vi) all claims in (c) with **explicit verification** that $40 \in N$ (min exists) and $50 \notin N$ (max does not exist); (vii) both interpretations.

Coherence rules — Question 3

- **Rounding at equilibrium:** $D(50) = 1148.6 \approx 1148.8$ — no penalty. Document the rounding.
- **Wrong sign in shortage/surplus:** deduct 2 pts. Grade $I(M)$ and $N(M)$ under coherence with wrong zones.
- **$I(M)$ correct, $N(M)$ incorrect:** deduct 2 pts. Grade length condition under coherence.
- **Wrong M from correct length setup:** no additional deduction beyond $N(M)$ error.
- **Argues $\min N$ does not exist** (the most common error): **deduct 3 pts.** This is the key conceptual distinction. The student must check whether the left endpoint belongs to N — it does, because the intervention condition is strict.
- **Correct min/max conclusions without verification:** deduct 1 pt per unsupported claim.
- **Omits economic interpretations:** deduct 1 pt per missing sentence.